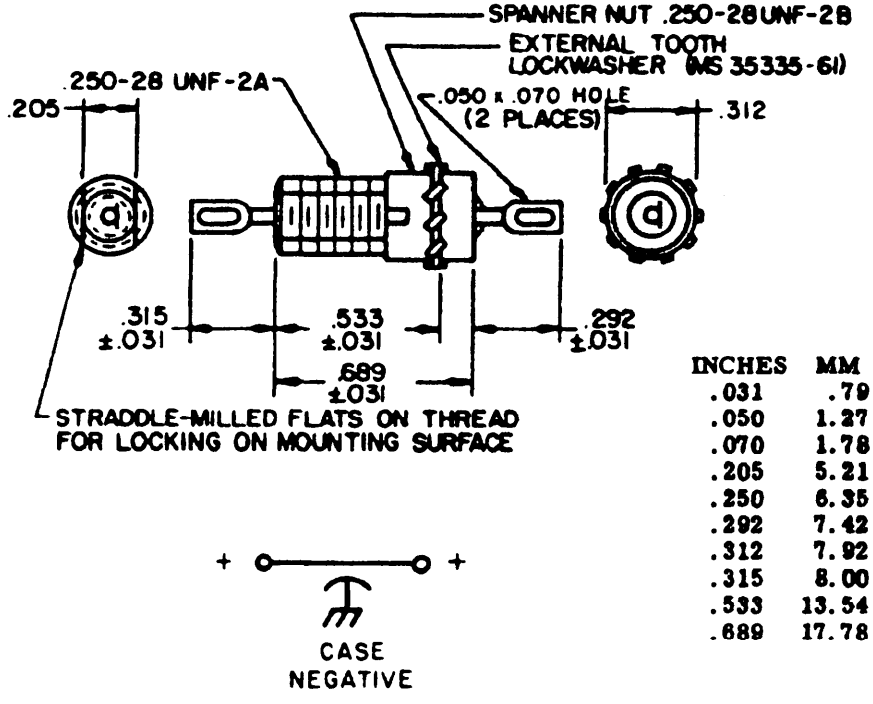
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, FEED THROUGH, ELECTROMAGNETIC INTERFERENCE SUPPRESSION, DC (HERMETICALLY SEALED IN METAL CASES)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-83439.



INCHES	MM
.031	.79
.050	1.27
.070	1.78
.205	5.21
.250	6.35
.292	7.42
.312	7.92
.315	8.00
.533	13.54
.689	17.78

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.010 (.25 mm).
4. Circuit diagram is for information only.
5. Mounting hardware (lockwasher and spanner nut) shall be supplied with capacitor.

FIGURE 1. Case dimensions and circuit configuration.

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TABLE I. Electrical characteristics.

Cap. value (uF)	DC voltage			DC leakage (in uA)			Minimum insertion loss (in dB) at frequency (in MHz)									Part No. M83439/05-
	Rated 85°C	Surge 85°C	Derated 125°C	25°C	85°C	125°C	0.014	0.02	0.05	0.10	1.00	10.0	100	400	1000	
60	6	8	4	7.2	72.0	90.0	28	29	31	33	39	41	42	48	54	007
39	10	13	7	7.8	78.0	97.5	24	25	27	28	33	35	40	46	52	001
22	15	20	10	6.6	66.0	82.5	22	23	25	27	32	34	38	45	50	002
15	20	26	13	6.0	60.0	75.0	22	23	25	27	32	34	38	45	50	003
6.8	35	46	23	4.8	47.6	59.5	19	20	22	23	27	29	36	43	48	004
4.7	50	65	33	4.7	47.0	58.8	19	20	22	23	27	29	36	43	48	005
3.9	75	97	50	5.9	58.5	73.1	17	18	20	22	25	27	29	36	41	006

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REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figure 1.
Material:
Case and terminals: Metal.
Finish: Corrosion-resistant.
Dielectric: Tantalum (solid electrolyte).
Temperature range: -55° to +125°C.
Rated voltage: See table I. Full rated voltage from -55° to +85°C, derate linearly to +125°C.
Current rating: 5 amperes at +85°C; 2 amperes at +125°C.

Capacitance (nominal): See table I.
Test frequency: 120 Hz.
Magnitude of polarizing voltage: 2.2 volts for all ac measurements. The magnitude of the ac voltage shall be limited to 1.0 volt rms.
Capacitance tolerance: ±20 percent.

Flashpoint of impregnant or filling compound: Not applicable.

Seal: Method 112 of MIL-STD-202, test-condition letter B.

Terminal strength: Method 211 of MIL-STD-202, test-condition letter A.
Applied force: 3 pounds $^{+3}_{-0}$ ounces.

DC leakage (maximum): See table I (+25°C).

Dissipation factor (maximum):
At +25°C, +85°C, and +125°C: 6 percent.

Insertion loss: See table I (for group A inspection, testing need only be done at 400 MHz and 1 MHz).

Overload: Not applicable.

Thermal shock and immersion:
Thermal shock: Method 107 of MIL-STD-202, test-condition letter A.
Immersion: Method 104 of MIL-STD-202.
Measurements after final cycle - DC leakage, capacitance, and dissipation factor shall remain within initial requirements.

Shock (specified pulse): Method 213 of MIL-STD-202, test-condition letter I; rated voltage applied.

Vibration, high frequency: Method 204 of MIL-STD-202, test-condition letter B; rated voltage applied during test.

Moisture resistance: Method 106 of MIL-STD-202.
Polarization voltage - Not applicable.
Measurements after test - DC leakage, capacitance, and dissipation factor shall remain within initial requirements.

Life at +85°C: Method 108 of MIL-STD-202, test-condition letter F; rated voltage applied.
Measurements after exposure:
DC leakage shall not exceed +85°C value of table I.
Capacitance shall change not more than ±10 percent of initial value.
Dissipation factor shall meet initial requirement.
Insertion loss (abbreviated) shall meet value of table I at 400 MHz and 1 MHz.

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Life at +125°C: Method 108 of MIL-STD-202, test-condition letter D; derated voltage applied.
Measurements after test:
DC leakage shall not exceed +125°C value of table I.
Capacitance shall not have changed by more than ±10 percent of initial value.
Dissipation factor shall meet the initial requirement.

1/ Surge voltage: Capacitors shall be subjected to 1,000 cycles of the applicable surge voltage shown in table I. The ambient temperature during cycling shall be +85°C. Each cycle shall consist of a 30-second surge voltage application followed by a 30-second discharge period. Voltage application shall be made through a resistor of 33 ohms. The tolerance of the resistor shall be ±5 percent. Each surge voltage cycle shall be performed in such a manner that the capacitor is shorted terminal-to-terminal through a copper bar, or an equivalent low resistance at the end of the 30-second application. An alternate method of shorting the capacitors is discharge through the same resistance that is utilized for charging. After the final cycle, the capacitors shall be stabilized and the dc leakage, capacitance, and dissipation factor shall be measured.
Measurements after exposure:
DC leakage shall not exceed the +25°C value shown in table I.
Capacitance shall not have changed by more than ±2 percent of the initial value.
Dissipation factor shall meet the initial requirement.

Marking: In accordance with MIL-C-83439, except part number only (see table I).

1/ For qualification inspection, six sample units shall be tested as the last test; 1 failure allowed. For quality conformance inspection, six sample units shall be tested as the last test in group B; 1 failure allowed.

Review activities:
Air Force - 11, 85
DSA - ES

Preparing activity:
Air Force - 11

Agent:
DSA - ES

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